

Software Requirements Specification

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Section 1: Purpose

The Purpose of this software is to create an easier way to create fundraisers for the local area. The purpose of this document is to specify high-level requirements that will be implemented in our project.

Section 2: Scope

The software product that will be produced is called The Fundraising Website. The Fundraising Website will be able host fundraising events and will be able to collect donations. The Fundraising Website's goal is to create a space for people in the local area to support people in the community.

Section 3: Product Perspective

This fundraising website will function as an independent web application. It is not a part of a larger system and does not require additional software installations. Users will access the platform through a web browser and the data will be stored in a database that keeps track of fundraisers and donation information. That platform exists to give school school groups and local organizations a simple system for creating and managing fundraising events.

Section 4: Product Functions

The main functions of the fundraising website include:

- Creating and editing fundraiser pages
- Displaying the fundraiser goal and total money collected
- Allowing users to submit donation entries
- Automatically updating donation totals and progress
- Showing donation history and basic donor information (if provided)

The goal is to keep all features straightforward so users can easily track and manage fundraising efforts.

Section 5: User Characteristics

The intended group of people for this application are local high school students and people in the nearby area.

Section 6: Limitations

Some limitations that this application faces is that it needs to be secure, it needs to be able to update quickly, and it needs to be reliable.

Section 7: Assumptions and Dependencies

We assume users will access the system from a supported web browser and have reliable internet access. We also assume that donation entries will be input accurately by users, since the system does not automatically verify real financial data. The website relies on a functioning database and web hosting environment to operate correctly.

Section 8: Apportioning of Requirements

Essential features like creating fundraisers, entering donations, displaying totals will be implemented first. Additional features like automated emails, user profiles, or online payment integration may be considered for future development but are not required for the initial version.

Section 9: Specified Requirements

- Ability to create and view fundraiser pages
- Input fields for donation amount and optional donor name
- Automatic calculation and display of total funds raised
- Visible progress toward a fundraising goal
- Stored data should remain available after closing the browser.

Section 10: External Interfaces

Users will interact with the system through a web interface. Inputs include text fields for fundraiser information and donation amounts. Output includes displayed totals, fundraiser details and updated progress visuals. No external payment systems or APIs are connected at this stage.

Section 11: Functions

The system will validate donation inputs, store the data, update the data. Update totals and refresh the fundraiser displays. It will also prevent invalid entries such as negative donation values. If incorrect data is entered, the system will present a clear message to the user instead of crashing.

Section 12: Usability Requirements

Usability can be defined as being defined as having a low number of errors per thousand lines of code. Quality can be defined as having the application being able to do its main goal with few errors and how often users are satisfied.

Section 13: Performance Requirements

This application should be able to support 100 people at once and should be able 100% of the time to take payments and create fundraisers.

Section 15: Design Constraints

The software must run in a standard web browser and follow basic web design practices. Project development will follow the class's software process requirements, including version control and scheduled reviews. No special hardware or external software will be required.

Section 16: Standards Compliance

In our SQAP, we outlined a series of quantifiable standards that our software will be tested on in order to ensure quality. These standards will be checked through both implementation as well as demos to prove our project's quality.

Section 17: Software System Attributes

Reliability - The website should be able to take payments in under three seconds 90% of the time.

Availability - This website should be able to be accessed at any time

Security - This website should be able to maintain data privacy, keep logs of payments, and create secure accounts(with 2FA)

Section 18: Verification

We will be implementing unit tests as we develop to ensure quality. These quantifiable metrics include correct amounts of money being transferred, rounding accuracy, load-time, as well as the number of donations allowed in a set amount of time.

Section 19: Supporting Information

Creating fundraisers specifically for a local area can be very difficult so, with this application we are making it easier for people in the local community to create fundraisers.